

Joint limit of the normalised chart integrals for deterministic amplitudes

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Abstract

Astra #13 unit u158. In §4.3 only the phase ξ_n is random; the amplitudes $\eta_\varphi = \varphi \cdot \text{prior} \cdot \text{Jac}$ and $\eta_1 = \text{prior} \cdot \text{Jac}$ are deterministic. We model this by `withY y`, the map on the coefficient-pair space that keeps the phase data and replaces the amplitude data by a fixed array y (1-Lipschitz). For equal starting exponents p the leading exponent has no lower terms, and the pair of normalised chart integrals $(Z_{N_n}[\varphi]/s_n, Z_{N_n}[1]/s_n)$ with $s_n = N_n^{-p} \log N_n$ converges jointly in distribution to $(A_p(X; y_\varphi), A_p(X; y_1))$ whenever the random phase data converge in distribution (`tendstoInDistribution_normA_pair`). Zero sorry/axiom.

1 The things formalised

```
def withY ( ) (h) (y) (hy : WSummable y) (a : CoeffPair) : CoeffPair :=
  ofPair h (toX a) y (wsumnable_toX h a) hy
theorem toX_withY / toY_withY / withY_apply_inl / withY_apply_inr
theorem norm_withY_sub_le : ||withY ... a - withY ... a'|| ||a - a'||
theorem lipschitzWith_withY / continuous_withY / measurable_withY
theorem lowerPart_leading_eq_zero (h h k k) (hk hk) (p T) (hp hp) (A B) (N) :
  lowerPart (polesBelowGen h h k k p p T) A B p N = 0
theorem tendstoInMeasure_prodMk_zero ( ) (f g) (hf : TendstoInMeasure f l (fun _ => 0))
  (hg) :
  TendstoInMeasure (fun n => (f n , g n )) l (fun _ => (0, 0))
theorem tendstoInMeasure_normA_withY_sub ... (hpT : p < 2 * T) (y) (hy) (X) (Z) (hX)
  (Nseq) (hN) :
  TendstoInMeasure (fun n => normA ... p (Nseq n) (withY ... y hy (X n ))
    - coeffA ... p (withY ... y hy (X n ))) l (fun _ => 0)
theorem tendstoInDistribution_normA_pair ... (y y) (hy hy) (X) (hXm) (Z) (hX)
  (Nseq) (hN) :
  TendstoInDistribution (fun n => (normA ... p (Nseq n) (withY ... y ... (X n )),
    normA ... p (Nseq n) (withY ... y ... (X n )))) l
    (fun _ => (coeffA ... p (withY ... y ... (Z )), coeffA ... p (withY ... y ... (Z ))))
    (fun _ => ) '
```

2 Mathematics

`withY y a` has the same inl-coordinates as a and fixed inr-coordinates, so $\|\text{withY } y \, a - \text{withY } y \, a'\| \leq \|a - a'\|$ termwise. The leading exponent $p = \min \Lambda$ has empty lower part. The two normalised chart integrals are each the corresponding coefficient plus an error that tends to zero in probability (the argument of units 152–153 applied to the pushed-forward data `withY y o Xn`, which converge in

distribution by the continuous mapping theorem and are therefore norm-bounded in probability); the pair of errors tends to zero in probability, and the pair of coefficients converges jointly by the continuous mapping theorem applied to the continuous map $a \mapsto (A_p(\text{withY } y_\varphi a), A_p(\text{withY } y_1 a))$.

3 Paper-facing meaning

This is the joint numerator/denominator convergence that `cor:empirical_expectation`'s division needs, in the paper's own setting (deterministic amplitudes, random phase). With the quotient theorem of unit 157 and the positivity of unit 154 the chart posterior limit follows in the next unit, including Astra #13's observation that the common factor $\int s^{p-1} e^{-\beta s^2 + \beta s x_{00}} ds$ cancels, so that $Z_n[\varphi]/Z_n[1] \Rightarrow y_{\varphi,00}/y_{1,00}$.

4 Lean notes

- Coordinates of `ofPair` are exposed by `change` (the underlying function is a `Sum.elim`); `lp.coeFn_sub` then `Pi.sub_apply` for differences.
- `LipschitzWith.of_dist_le_mul` with `NNReal.coe_one`.
- `Prod.mk_sub_mk` and `Prod.norm_def` ($\|(a, b)\| = \max(\|a\|, \|b\|)$) reduce the pair convergence in measure to the components via `measure_union_le`.
- For `ENNReal`, `zero_le` takes its argument implicitly; unused hypotheses are underscored and the unused section variable is `omitted`.